Features, Labels, Targets

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Features, Labels, Targets before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a **label**led second line to correct your explanation.

The ML Lifecycle

Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Features, Labels, Targets trace and record the first step you can explain without help.

Use W2: Trace the **feature contract** for **Feature**s **label**s and **prediction time**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Features, Labels, Targets explanation against the supplied facts before discussing it.

Use the worked case for **Feature**s **label**s and **prediction time**. Proposed explanation: “Every stored column is a **feature**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Audit a new task independently

A fictional robot predicts at dispatch whether a parcel trip will take more than 10 minutes.

Fields are route_steps and queue_at_dispatch captured at dispatch, trip_id for tracking, arrival_minutes measured after arrival, and slow = 1 when arrival_minutes > 10. Specify **input**s, order, **target** and exclusions.

For a past trip with steps 30, queue 2, arrival 10, give **feature**s and **label**. For a new trip with steps 24 and queue 0, give **feature**s and explain whether slow is known.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Features, Labels, Targets target to a trace and a justified conclusion.

A table has leaf length, edge type and verified species. A model predicts species using all three columns. Identify leakage and a legitimate **input** set.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Features, Labels, Targets case and show the deciding evidence.

Rows contain leaf length, edge type and healthy/spotted **label**.

Identify **feature**s, **target** and **prediction**-time **input**s.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A row contains leaf_length_cm=8, leaf_width_cm=3 and species=fern. For predicting species, identify two **feature**s and the **target**.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 2 Test and reflect

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	

Test record	My evidence
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Features, Labels, Targets adds to Responsible Plant Classifier and keep the evidence.

Define the Plant Classifier **target** and a first **feature** list; record units, allowed values, collection method, and availability time.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.

- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A table has leaf length, edge type and verified species. A model predicts species using all three columns. Identify leakage and a legitimate **input** set.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Assign the roles

> Assign the roles. Apply the supplied Features, Labels, Targets facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Features, Labels, Targets", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For the worked loan table, identify the two candidate **feature**s, the **target label** and the tracking field. Explain why days_to_return is not a predictor.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the feature contract

> Trace the feature contract. Apply the supplied Features, Labels, Targets facts and keep your reasoning.

A completed loan has pages 96, queue_at_checkout 4 and days_to_return 14. Give the ordered **feature** list and **target**. Then state what changes if it took 15 days instead.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Repair a misleading score

> Repair a misleading score. Apply the supplied Features, Labels, Targets facts and keep your reasoning.

A model has every test **prediction** correct after using pages and days_to_return. A learner says a separate test set makes this valid. Explain the flaw and repair the **input**s.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Audit a new task independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Audit a new task independently. Apply the supplied Features, Labels, Targets facts and keep your reasoning.

A fictional robot predicts at dispatch whether a parcel trip will take more than 10 minutes.

Fields are route_steps and queue_at_dispatch captured at dispatch, trip_id for tracking, arrival_minutes measured after arrival, and slow = 1 when arrival_minutes > 10. Specify **input**s, order, **target** and exclusions.

For a past trip with steps 30, queue 2, arrival 10, give **feature**s and **label**. For a new trip with steps 24 and queue 0, give **feature**s and explain whether slow is known.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Handle an unavailable input

> Handle an unavailable input. Apply the supplied Features, Labels, Targets facts and keep your reasoning.

For the robot task, a dataset stores queue_now at export time but has no saved queue_at_dispatch. Can you rename that field to make it usable? State a defensible next step.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Features, Labels, Targets

> Choose features that exist when a plant prediction must be made. Explain the labelled step before checking the changed condition.

Choose **feature**s that exist when a plant **prediction** must be made.

Predict next-week wilting at today's photo capture. Available fields: today_light, today_soil_moisture, plant_id; next_week_wilt is the **target** recorded later.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/2

Features, Labels, Targets

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Mark prediction time

The decision occurs today, before next week's outcome.

2. Select inputs

Today's light and soil readings are available features; plant_id identifies the plant.

3. Keep target separate

next_week_wilt supplies the training label, not an input available today.

CHANGE ONE THING

A table adds tomorrow_soil_moisture as an input.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why can a historical table contain a field that is an invalid **input**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Choose **feature**s that exist when a plant **prediction** must be made.

Practice 1: explain the deciding evidence

Which **label**led step matches this detail? next_week_wilt supplies the **training label**, not an **input** available today.

1. Select **input**s

2. Mark **prediction time**

3. Keep **target** separate

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Keep **target** separate next_week_wilt supplies the **training label**, not an **input** available today.

Read the picture in order. next_week_wilt supplies the **training label**, not an **input** available today. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A table adds tomorrow_soil_moisture as an **input**. Which conclusion is supported by the supplied facts?

1. Reject it for this **prediction**-time contract.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reject it for this **prediction**-time contract. Tomorrow's measurement is unavailable when today's **prediction** must be issued.

Repair a missing building block

Review The ML Lifecycle (ai-4/1). Goal: Distinguish **fitting** a rule from applying a fixed rule. Short practice: Should the team silently retune on the changed test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, **label** your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

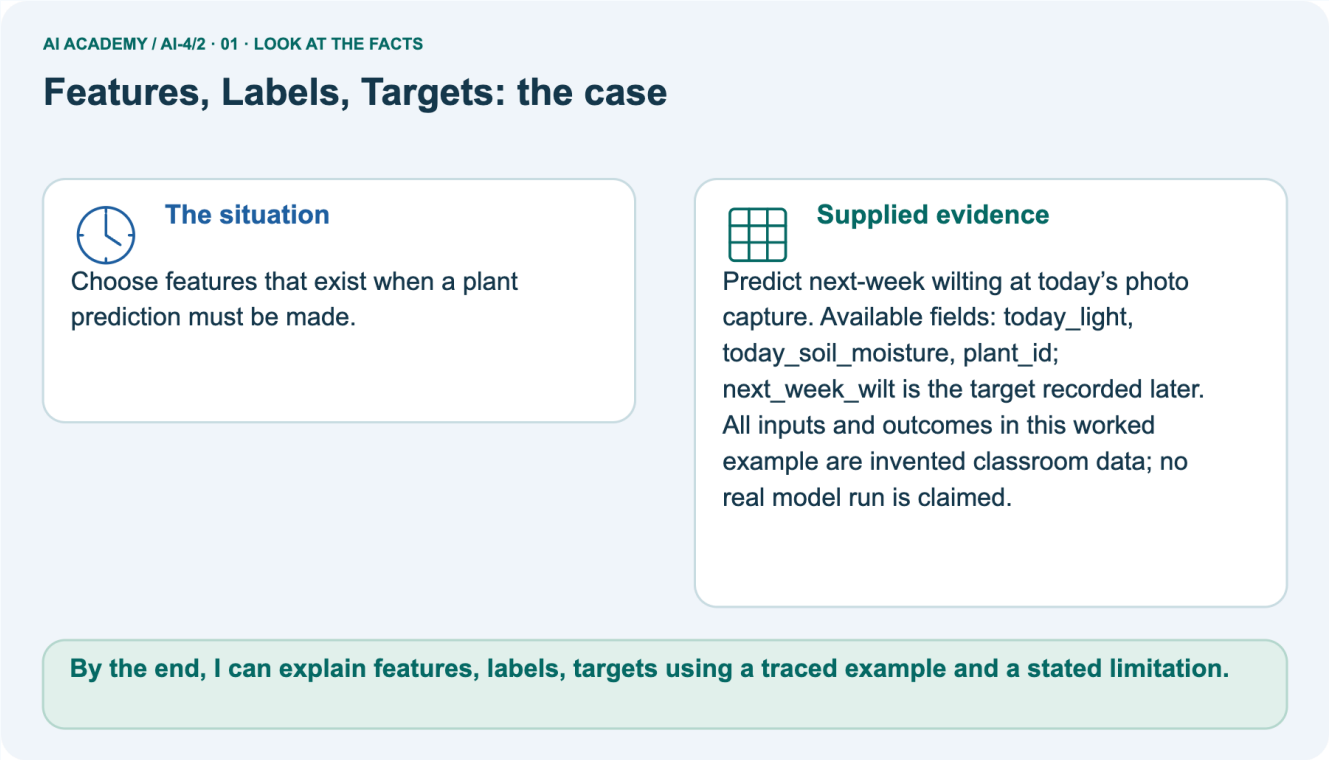
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Features, Labels, Targets picture from its labels, then record a relevant limit.

01. Features, Labels, Targets: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

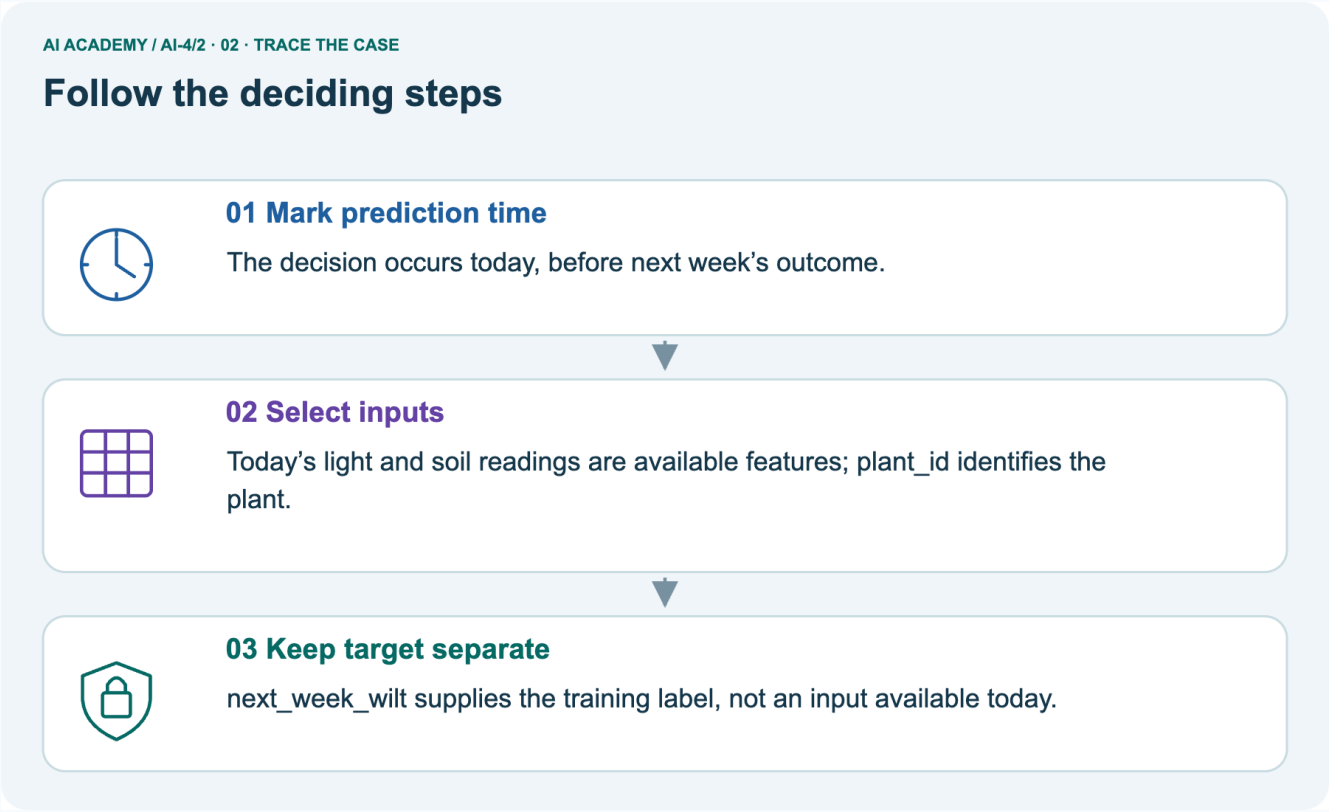
Choose **feature**s that exist when a plant **prediction** must be made. Supplied facts: Predict next-week wilting at today's photo capture.

Available fields: today_light, today_soil_moisture, plant_id; next_week_wilt is the **target** recorded later. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



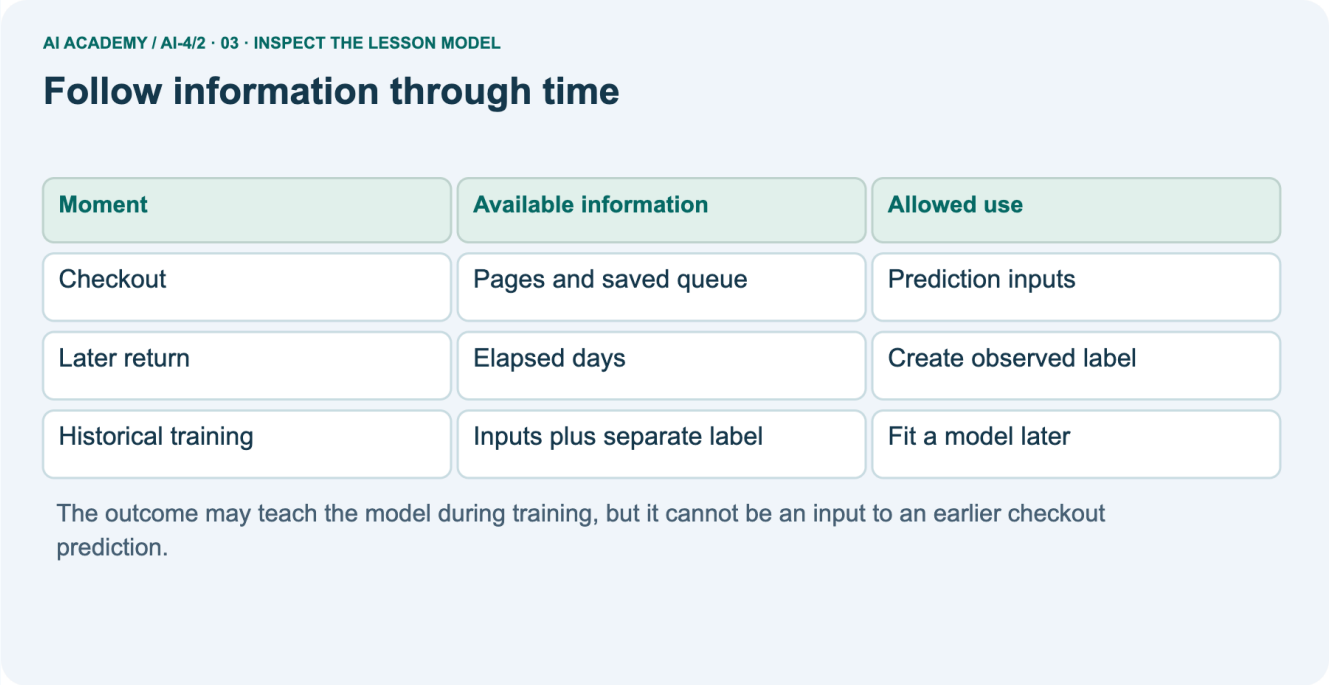
Follow the labelled picture with the matching explanation. Open the vector diagram

Mark **prediction time**: The decision occurs today, before next week's outcome. → Select **input**s: Today's light and soil readings are available **feature**s; plant_id identifies the plant. → Keep **target** separate: next_week_wilt supplies the **training label**, not an **input** available today.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow information through time



AI ACADEMY / AI-4/2 · 03 · INSPECT THE LESSON MODEL

Follow information through time

Moment	Available information	Allowed use
Checkout	Pages and saved queue	Prediction inputs
Later return	Elapsed days	Create observed label
Historical training	Inputs plus separate label	Fit a model later

The outcome may teach the model during training, but it cannot be an input to an earlier checkout prediction.

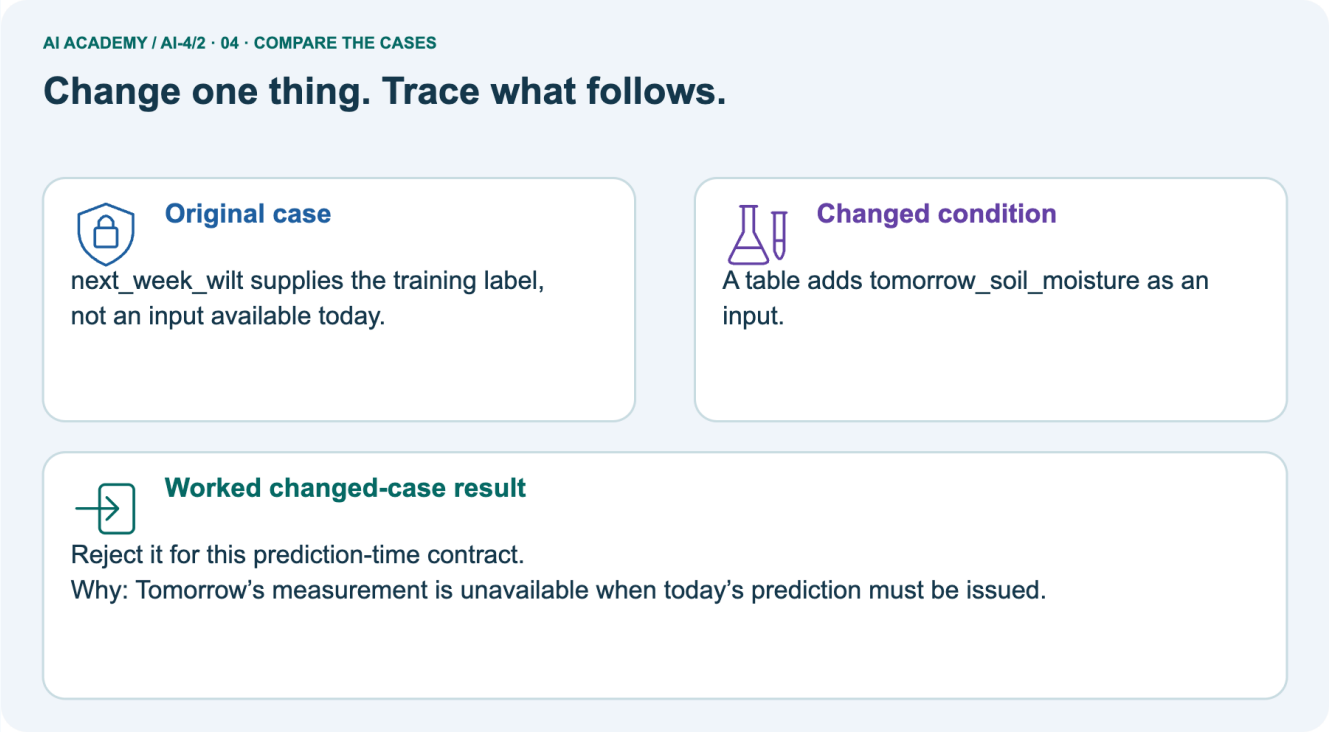
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow information through time. Checkout | Pages and saved queue | **Prediction input**s. Later return | Elapsed days | Create observed **label**. Historical **training** | **Input**s plus separate **label** | Fit a model later

Read the **label**s and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

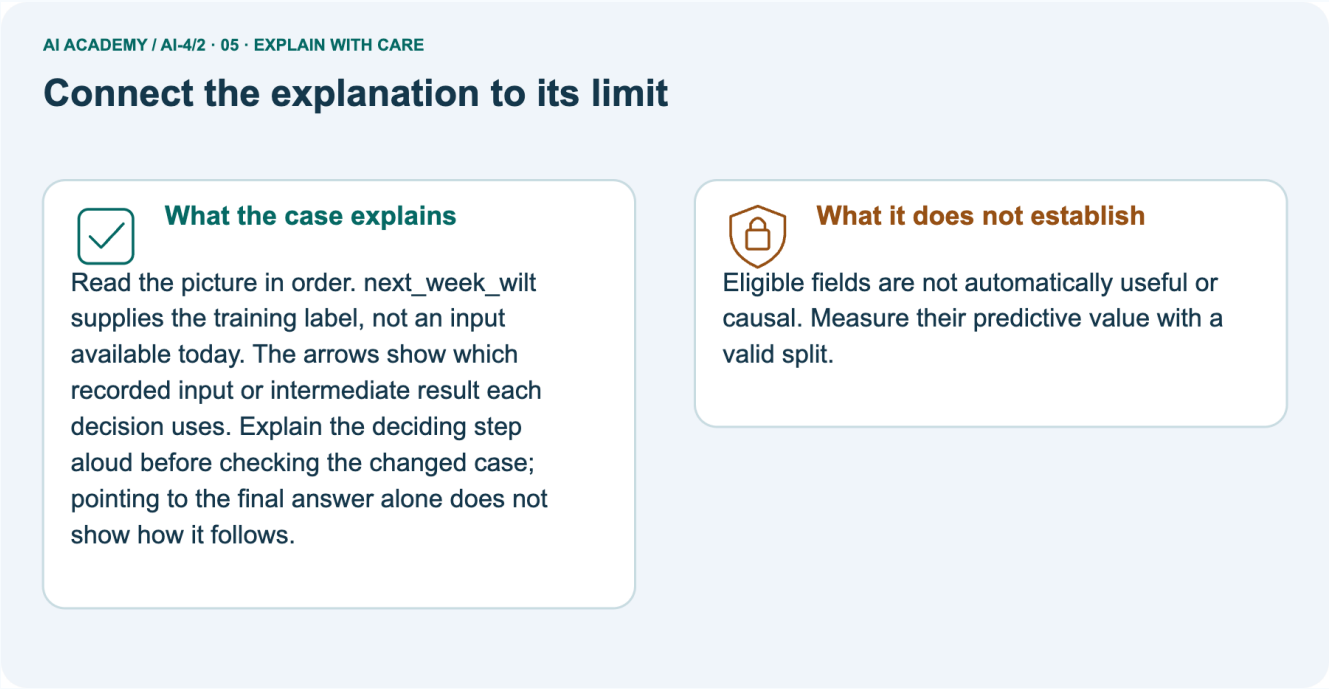
Original case: next_week_wilt supplies the **training label**, not an **input** available today. Changed condition: A table adds tomorrow_soil_moisture as an **input**. Expected result: Reject it for this **prediction**-time contract.

Reason: Tomorrow's measurement is unavailable when today's **prediction** must be issued.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. next_week_wilt supplies the **training label**, not an **input** available today. The arrows show which recorded **input** or intermediate result each decision uses.

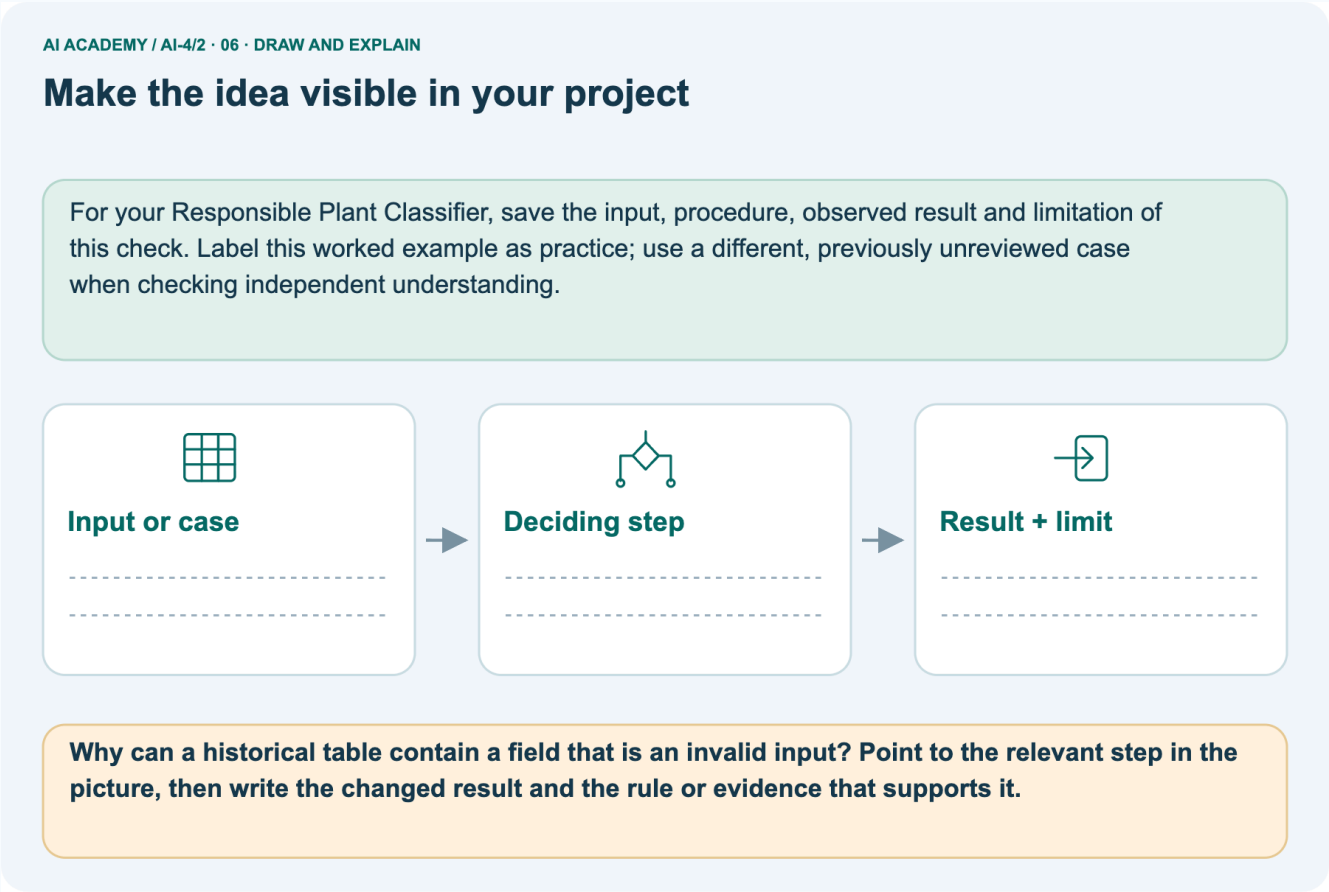
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Eligible fields are not automatically useful or causal.

Measure their predictive value with a valid split.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can a historical table contain a field that is an invalid **input**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.